

# Linear Algebra for AI

## Final Exam 2023

①  $x = (x_1, x_2, x_3, x_4, x_5)$ ,  $T: \mathbb{R}^5 \rightarrow \mathbb{R}^4$  linear is given by  $T = \begin{bmatrix} x_1 + 2x_3 + x_4 \\ -x_1 + x_2 + x_3 + x_5 \\ -x_2 - x_3 + 2x_4 + x_5 \\ x_1 + x_2 - x_4 - x_5 \end{bmatrix}$ .

a) What is the standard matrix of  $T$ ?

Let's call it  $A$ .  $A = \begin{bmatrix} 1 & 0 & 2 & 1 & 0 \\ -1 & 1 & 1 & 0 & 1 \\ 0 & -1 & -1 & 2 & 1 \\ 1 & 1 & 0 & -1 & -1 \end{bmatrix}$ .

b) Basis for  $\text{Col} A$ .

It will be made of the pivot columns of  $A$ .  $A \sim \begin{bmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 3 & 1 & 1 \\ 0 & -1 & -1 & 2 & 1 \\ 0 & 1 & -2 & -2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 3 & 2 \\ 0 & 0 & -5 & -3 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 3 & 1 & 1 \\ 0 & 0 & 2 & 3 & 2 \\ 0 & 0 & 0 & 9 & 6 \end{bmatrix}$

So a basis for  $\text{Col} A$  is  $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \\ 1 \end{bmatrix} \right\}$ .

c) Basis for  $\text{Nul} A$ .

We must find the solutions for  $Ax = 0$ . Picking up from the previous reduction:  $[A|0] \sim \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 1 & 1 & 0 \\ 0 & 0 & 2 & 3 & 2 & 0 \\ 0 & 0 & 0 & 9 & 6 & 0 \end{bmatrix} \sim$   
 $\sim \begin{bmatrix} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 3 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 & 0 & -2/3 & 0 \\ 0 & 1 & 3 & 0 & 1/3 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2/3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & -2/3 & 0 \\ 0 & 1 & 0 & 0 & 1/3 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2/3 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 = \frac{2}{3}x_5 \\ x_2 = -\frac{1}{3}x_5 \\ x_3 = 0 \\ x_4 = -\frac{2}{3}x_5 \end{cases} \rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = x_5 \begin{bmatrix} 2/3 \\ -1/3 \\ 0 \\ -2/3 \\ 1 \end{bmatrix}$ . So a basis for  $\text{Nul} A$  is  $\left\{ \begin{bmatrix} 2/3 \\ -1/3 \\ 0 \\ -2/3 \\ 1 \end{bmatrix} \right\}$ .

② Let  $B = \begin{bmatrix} -1 & 2 & -1 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix}$ .

a) Give the LU factorization.

Start reducing  $B$ .  $\begin{bmatrix} -1 & 2 & -1 \\ 1 & 0 & 1 \\ -2 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & -1 \\ 0 & 2 & 0 \\ 0 & -3 & 1 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = U$

And  $L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -3/2 & 1 \end{bmatrix}$ .

b)  $B^{-1}$ ?

If it's invertible then  $[B|I] \sim [I|B^{-1}]$ .

$\begin{bmatrix} -1 & 2 & -1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ -2 & 1 & -1 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} -1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & -3 & 1 & -2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 1 & -1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & -1 & 3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & -1/2 & 3/2 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 0 & -1/2 & -3/2 & -1 \\ 0 & 1 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & -1/2 & 3/2 & 1 \end{bmatrix} \sim$   
 $\sim \begin{bmatrix} 1 & 0 & 0 & 1/2 & -1/2 & -1 \\ 0 & 1 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & -1/2 & 3/2 & 1 \end{bmatrix}$ . So  $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & -1 \\ 1/2 & 1/2 & 0 \\ -1/2 & 3/2 & 1 \end{bmatrix}$ .

c) Let  $x = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$ . Find  $[x]_B$ , where  $B$  is made of the columns of  $B$ .

We know  $x = B[x]_B$ , so  $[x]_B = B^{-1}x = \begin{bmatrix} 1/2 & -1/2 & -1 \\ 1/2 & 1/2 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} -3/2 \\ 1/2 \\ 3/2 \end{bmatrix}$ .

③ Let  $H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0 \right\}$ .

a) Prove that  $H$  is a subspace of  $\mathbb{R}^3$ .

We must check three things:

(i)  $\vec{0} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \in H$  since  $0+0+0=0$ .

(ii) Let  $u, v \in H$ , with  $u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$  and  $v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$ , then  $u_1+u_2+u_3 = v_1+v_2+v_3 = 0$ . Then  $u+v = \begin{bmatrix} u_1+v_1 \\ u_2+v_2 \\ u_3+v_3 \end{bmatrix}$ , and  $(u_1+v_1) + (u_2+v_2) + (u_3+v_3) = (u_1+u_2+u_3) + (v_1+v_2+v_3) = 0+0=0$ , so  $u+v \in H$ .

(iii) Let  $u \in H$  like before and  $c \in \mathbb{R}$ . Then  $cu = \begin{bmatrix} cu_1 \\ cu_2 \\ cu_3 \end{bmatrix}$ , and  $cu_1+cu_2+cu_3 = c(u_1+u_2+u_3) = c \cdot 0 = 0$ . So  $cu \in H$ .

With this we conclude  $H$  is a subspace of  $\mathbb{R}^3$ .

b) Find a basis of  $H$ .

Since if  $x \in H$  then  $x_1+x_2+x_3=0$ ,  $x_1 = -x_2-x_3$  has two free variables, so  $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$  and a basis is  $B = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ .

④ Let  $A = \begin{bmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 2 \end{bmatrix}$ .

a) Show that  $v_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$  is an eigenvector of  $A$  and find  $\lambda_1$ .

$Av_1 = \begin{bmatrix} -3 \\ 1 \\ -1 \end{bmatrix} = -1 \cdot v_1$   
 $\lambda_1 = -1$

b) Give all eigenvalues and give a basis for every eigenspace.

We know by a)  $\lambda_1 = -1$ .  $\det(A - \lambda I) = (3-\lambda) \begin{vmatrix} 2-\lambda & 3 \\ 1 & 2-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 3 \\ 1 & 2-\lambda \end{vmatrix} + \begin{vmatrix} 1 & 2-\lambda \\ 1 & 3 \end{vmatrix} = (3-\lambda)((2-\lambda)^2 - 9) - 2(2-\lambda-3) + 3-2+\lambda = (3-\lambda)(4-4\lambda+\lambda^2-9) - 2(-1-\lambda) + 1+\lambda = (3-\lambda)(-5-4\lambda+\lambda^2) + 2+2\lambda+1+\lambda = -15-12\lambda+3\lambda^2+5\lambda+4\lambda^2-\lambda^3+3+3\lambda = -\lambda^3+7\lambda^2-4\lambda-12$

Since  $\lambda_1 = -1$  is an eigenvalue, we can divide the result by  $(\lambda+1)$  to get the remaining factors.

$$\begin{array}{r} -\lambda^3 + 7\lambda^2 - 4\lambda - 12 \quad | \lambda + 1 \\ -(\lambda^3 - \lambda^2) \quad \quad \quad | -\lambda^2 + 8\lambda - 12 \\ \hline 0 \quad 8\lambda^2 - 4\lambda - 12 \\ \quad - (8\lambda^2 + 8\lambda) \quad \quad \quad | \\ \hline 0 \quad -12\lambda - 12 \\ \quad - (-12\lambda - 12) \quad \quad \quad | \\ \hline 0 \end{array}$$

$-\lambda^2 + 8\lambda - 12 = 0$

$\lambda = \frac{-8 \pm \sqrt{64 - 4(-1)(-12)}}{-2 \cdot 1}$

$= \frac{-8 \pm \sqrt{64 - 48}}{-2} = \frac{-8 \pm 4}{-2} = \begin{cases} \lambda_2 = 2 \\ \lambda_3 = 6 \end{cases}$

So the eigenvalues are  $\lambda_1 = -1, \lambda_2 = 2, \lambda_3 = 6$ . Since they are all different with multiplicity 1 and  $A$  is  $3 \times 3$ , the eigenspaces will have dimension 1, hence for  $\lambda_1, \{v_1\}$  will be a basis for the eigenspace. Let's find them for the rest.

$A - \lambda_2 I = A - 2I = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 3 \\ 1 & 3 & 0 \end{bmatrix}$ . Solve  $(A - \lambda_2 I)x = 0$ .

$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 3 & 0 \\ 1 & 3 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 + 3x_3 = 0 \\ x_2 - x_3 = 0 \end{cases} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \stackrel{v_2}{\parallel}$

So a basis for the eigenspace is  $\left\{ \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \right\}$ .

Now  $A - \lambda_3 I = A - 6I = \begin{bmatrix} -3 & 2 & 1 \\ 1 & -4 & 3 \\ 1 & 3 & -4 \end{bmatrix}$ . Let's solve  $(A - \lambda_3 I)x = 0$ .

$\begin{bmatrix} -3 & 2 & 1 & 0 \\ 1 & -4 & 3 & 0 \\ 1 & 3 & -4 & 0 \end{bmatrix} \sim \begin{bmatrix} -3 & 2 & 1 & 0 \\ 0 & -10 & 10 & 0 \\ 0 & 11 & -11 & 0 \end{bmatrix} \sim \begin{bmatrix} -3 & 2 & 1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} -3 & 0 & 3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{cases} x_1 = x_3 \\ x_2 = x_3 \end{cases} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \stackrel{v_3}{\parallel}$

So a basis for the eigenspace is  $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ .

c) Is  $A$  diagonalizable? If so give  $P$  and  $D$  such that  $A = PDP^{-1}$ .

Since  $A$  is  $3 \times 3$  and we have 3 different eigenvalues, we know  $A$  is diagonalizable and  $P = [v_1 \ v_2 \ v_3]$  with  $D = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 6 \end{bmatrix}$ .

⑤ Let  $W = \text{Span} \left\{ \underset{w_1}{\begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}}, \underset{w_2}{\begin{bmatrix} -1 \\ 1 \\ -3 \\ 3 \end{bmatrix}}, \underset{w_3}{\begin{bmatrix} 2 \\ 0 \\ -4 \\ 2 \end{bmatrix}} \right\}$

a) Find an orthogonal basis for  $W$ .

We would like to apply the Gram-Schmidt process to find an orthogonal basis. But first we need to check that  $\{w_1, w_2, w_3\}$  is a basis, that is, they are linearly independent.

$$\begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & 0 \\ 1 & -3 & -4 \\ -1 & 3 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 0 & 2 \\ 0 & -2 & -6 \\ 0 & 2 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 4 \\ 0 & -2 & -6 \\ 0 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 4 \\ 0 & 0 & -2 \\ 0 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

We have three pivot columns, so  $w_1, w_2, w_3$  form a basis.

Let's then apply Gram-Schmidt to find  $\{v_1, v_2, v_3\}$ .

$$v_1 = w_1$$

$$v_2 = w_2 - \frac{w_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} -1 \\ 1 \\ -3 \\ 3 \end{bmatrix} - \frac{(-8)}{4} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ -3 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 \\ -2 \\ 2 \\ -2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

$$v_3 = w_3 - \frac{w_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{w_3 \cdot v_2}{v_2 \cdot v_2} v_2 = \begin{bmatrix} 2 \\ 0 \\ -4 \\ 2 \end{bmatrix} - \frac{(-4)}{4} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} - \frac{8}{4} \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ -4 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 2 \\ -2 \\ -2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$$

So an orthogonal basis is  $\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix} \right\}$ .

b) Find the best approximation of  $y = \begin{bmatrix} 1 \\ 1 \\ 3 \\ 4 \end{bmatrix}$  by vectors of  $W$ .

We know by the Best Approximation Theorem that such best approximation  $\hat{y}$  is the orthogonal projection of  $y$  onto  $W$ , and is given by

$$\hat{y} = \frac{y \cdot v_1}{v_1 \cdot v_1} v_1 + \frac{y \cdot v_2}{v_2 \cdot v_2} v_2 + \frac{y \cdot v_3}{v_3 \cdot v_3} v_3 = \frac{-2}{4} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + 0 v_2 + \frac{-4}{4} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1/2 \\ 1/2 \\ -1/2 \\ 1/2 \end{bmatrix} + \begin{bmatrix} -1 \\ 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3/2 \\ 1/2 \\ -1/2 \\ 3/2 \end{bmatrix}$$

⑥ Prove or disprove:

a) The map  $T((x_1, x_2)) = (x_1 + x_2, x_1 - x_2, x_2)$  is linear.

We must check the conditions for a map to be linear.

$$\begin{aligned} T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) &= T\left(\begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}\right) = \begin{bmatrix} (x_1 + y_1) + (x_2 + y_2) \\ (x_1 + y_1) - (x_2 + y_2) \\ x_2 + y_2 \end{bmatrix} = \begin{bmatrix} (x_1 + x_2) + (y_1 + y_2) \\ (x_1 - x_2) + (y_1 - y_2) \\ x_2 + y_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 + y_2 \\ y_1 - y_2 \\ y_2 \end{bmatrix} \\ &= T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) + T\left(\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}\right) \end{aligned}$$

$$T\left(r \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = T\left(\begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}\right) = \begin{bmatrix} rx_1 + rx_2 \\ rx_1 - rx_2 \\ rx_2 \end{bmatrix} = \begin{bmatrix} r(x_1 + x_2) \\ r(x_1 - x_2) \\ rx_2 \end{bmatrix} = r \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \\ x_2 \end{bmatrix} = r T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right)$$

So true.

b) If  $A$  is  $3 \times 4$ , then  $Ax = 0$  only has the trivial solution.

False, since we know that a matrix with more columns than rows will always yield free variables (since we'll always have at least one non-pivot column), giving as a result non-trivial solutions.

c) If all eigenvalues are zero, then  $A$  is zero.

False. Take  $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ . Since it's triangular, we can read the eigenvalues from the diagonal to be 0, but  $A \neq 0$ .

⑦ Consider the  $n \times n$  matrix

$$A_n = \begin{bmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ -1 & 0 & 3 & \dots & n-1 & n \\ -1 & -2 & 0 & \dots & n-1 & n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -2 & -3 & \dots & 0 & n \\ -1 & -2 & -3 & \dots & -n+1 & 0 \end{bmatrix}$$

Show that  $\det A_n = n!$ .

Add the first row to all the rest to get  
(determinant doesn't change under such row  
operation)

$$\begin{bmatrix} 1 & 2 & 3 & \dots & n-1 & n \\ 0 & 2 & 6 & \dots & 2(n-1) & 2n \\ 0 & 0 & 3 & \dots & 2(n-1) & 2n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & n-1 & 2n \\ 0 & 0 & 0 & \dots & 0 & n \end{bmatrix}$$

, which is triangular, so the  
determinant is the product of  
the diagonal entries, so  
 $\det A_n = n!$ .